

# Quantitative Portfolio Risk Modeling & Markowitz Frontier

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## 1. Objective and Statistical Foundation

The objective of this analysis is to model and optimize a financial portfolio across 30 equity assets utilizing historical closing prices over a 120-day horizon. The foundational parameters driving the Markowitz Modern Portfolio Theory (MPT) framework are the Expected Returns ( $\mu$ ) and the Covariance Matrix ( $\Sigma$ ).

Initial exploratory analysis identified TATAMOTORS as the highest yielding individual asset (Return Rate: 0.0070), albeit with a correspondingly high standard deviation ( $\sigma = 0.0624$ ). Strong positive correlations were observed between specific sector equities, notably BAJAJFINSV and BAJFINANCE ( $\rho \approx 0.84$ ).

## 2. Mathematical Optimization Model

To map the risk-return characteristics, a quadratic programming (QP) model was formulated using the Pyomo library and solved via the IPOPT interior-point optimizer. The core optimization problem minimizes portfolio variance subject to budgetary and target return constraints:

$$\begin{aligned} \text{Minimize: } & \sum_{i \in S} \sum_{j \in S} x_i x_j \Sigma_{ij} \\ \text{Subject to: } & \sum_{i \in S} x_i = M \quad (\text{Budget Constraint}) \\ & \sum_{i \in S} \mu_i x_i \geq R_{\text{target}} \times M \quad (\text{Minimum Return Constraint}) \\ & x_i \geq 0 \quad \forall i \in S \quad (\text{Long-only Constraint}) \end{aligned}$$

where  $S$  represents the set of 30 assets,  $x_i$  is the capital allocation to asset  $i$ ,  $M$  is the total investment capital, and  $R_{\text{target}}$  is the dynamically iterated target return rate.

## 3. Efficient Frontier and Benchmarking

By iteratively solving the constrained quadratic program across varying return thresholds, the Markowitz Efficient Frontier was mapped. This frontier represents the set of mathematically optimal portfolios that offer the highest expected return for a defined level of risk.

To contextualize the optimization, an Equal-Weight ( $1/N$ ) baseline strategy was evaluated. Distributing capital equally across all 30 assets yielded an expected return of 0.0043 with a standard deviation (risk) of 0.0273.

**Conclusion:** The Equal-Weight baseline strategy is sub-optimal. The optimization model demonstrated that constrained-risk strategies mapping the efficient frontier can achieve equivalent returns with significantly reduced variance ( $\sigma \approx 0.0208$ ), structurally outperforming the naive equal-weight allocation.